

8. A number matching game stores four winning numbers and compares them to a player's input of four numbers. 250 points are awarded for each number matched. For example, two matched numbers would be awarded 500 points.

The code below contains an error, as the player is always awarded 1000 points.

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```
Line 10  DECLARE winningNos INITIALLY [10, 14, 21, 33]
Line 11  DECLARE numMatches INITIALLY 0
Line 12  DECLARE points INITIALLY 0
Line 13  RECEIVE no1 FROM KEYBOARD
Line 14  RECEIVE no2 FROM KEYBOARD
Line 15  RECEIVE no3 FROM KEYBOARD
Line 16  RECEIVE no4 FROM KEYBOARD
Line 17  FOR index FROM 0 TO 3 DO
Line 18      IF no1 = winningNos[index] THEN
Line 19          SET numMatches TO numMatches + 1
Line 20      ELSE IF no2 = winningNos[index] THEN
Line 21          SET numMatches TO numMatches + 1
Line 22      ELSE IF no3 = winningNos[index] THEN
Line 23          SET numMatches TO numMatches + 1
Line 24      ELSE
Line 25          SET numMatches TO numMatches + 1
Line 26      END IF
Line 27  END FOR
Line 28  SET points TO 250 * numMatches
Line 29  SEND "You matched "& numMatches & " numbers and have
        won " & points TO DISPLAY
```

...



* X 8 1 6 7 6 0 1 0 6 *

8. (continued)

- (a) A breakpoint is set at Line 26 and the program is tested using the following four player numbers as input:

no1	no2	no3	no4
5	10	15	22

Complete the trace table to show the values stored when the breakpoint is activated on the first two iterations of the loop.

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Breakpoint	Variable	Value
1 st iteration	index	0
	winningNos[index]	
	numMatches	
2 nd iteration	index	1
	winningNos[index]	
	numMatches	

- (b) Explain, with reference to the code, why the number of matches always results in 4.

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[Turn over



* X 8 1 6 7 6 0 1 0 7 *